



The Return of the King

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Ever since its Conroe architecture took the world by storm, Intel, the largest CPU manufacturer in the world has been on an undeniable roll. AMD has been unable to hit back with anything concrete and competition has been lean. Intel has even gone as far as promising us something new in terms of processor architecture every two years or so. In fact, many people have made much of Intel's "tick-tock" development model: each tick represents a die-shrink and each tock denotes a major architectural change. Late last year, Intel delivered the latest tock in the form of the processor code-named Nehalem; finally given the marketable moniker of Core i7, with its processor family dubbed "Bloomfield".

Intel's i7 is the new kid in town. Sure it's fast; but it has its roots in the past. We tell you how and why...

We tested a Nehalem processor, the i7 965 Extreme, a few months back and compared it to the (then) fastest desktop processor in the world — the Intel Core 2 Extreme QX 9770 (3.2 GHz). Needless to say, the Core i7's performance was at least 15 per cent faster, 90 per cent of the time, in both real-world tests and synthetic benchmarks. Not only was the i7 significantly faster than any AMD Phenom processor, it was also noticeably faster than the fastest quad-core Intel Yorkfield processor around. When you consider that the Core 2 architecture isn't very old and Intel has been continuously churning faster Core 2 processors, Nehalem becomes something rather special.

One of the firsts for Intel in the desktop space is that Nehalem is a *native* quad-core. Earlier, only AMD offered native quad-cores with its Phenom X4 range. Intel's quad-core processors before the i7 were basically two dual-core CPUs slapped together on a single die; a fact that has been of great interest to AMD, as it has thumbed its nose rather regularly at what it has called a Jurassic design. The market is, however, more concerned with real-world performance rather than on-paper specifications and the Core architecture is much faster than anything AMD has been able to conjure up. And now the newcomer is out with promises of even more processing grunt. Of course, AMD isn't whittling sticks either — its Phenom 2 processors have just made an entry — but they have a very tough opponent to beat.

Scalability

Intel claims to have designed the Core i7 with scalability in mind and looking at its released block diagrams we have to agree with them. In fact everything about the i7 design is modular, and while the current Nehalem CPUs are native quad-cores we could just as easily see eight-core CPUs or even dual-cores with integrated graphics, in the future. The fact is that Nehalem is designed from the ground up to be flexible and Intel can easily add or remove cores and other components as per market needs and demands. Nehalem also resurrects an old trick — hyperthreading (HT) — which was dropped since the Pentium 4. Hyperthreading is Intel's name for simultaneous multi-threading (SMT), a process in which operations from more than one instruction thread can be processed by a single processor core.

A quad-core Nehalem CPU can, therefore, process eight threads, while an eight core Nehalem will actually process a previously unheard of 16 threads. This gives the i7 much better multi-threaded performance while keeping thermal envelopes and silicon requirements much lower than they would be, if additional cores were added instead.

Doing more with less, seems to be Intel's design motto for the i7. HT also means fewer